higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 2

(12041032)

12 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

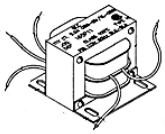
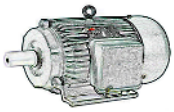
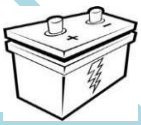
1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

1.1 Name TWO ways in which electricity can be distributed from the supplier to the consumer. (2)

1.2 Identify the electrical components in the table below. Write the name of the component and then draw the IEC symbol next to the relevant question number (1.2.1–1.2.10) in the ANSWER BOOK.

COMPONENT	NAME OF COMPONENT	IEC SYMBOL
	1.2.1	1.2.2
	1.2.3	1.2.4
	1.2.5	1.2.6
	1.2.7	1.2.8
	1.2.9	1.2.10

(10)

1.3 The reading recorded in an insulation resistance test between conductors was 0,6 MΩ.

Is this reading acceptable according to SANS 10142-1? Motivate your answer. (2)

1.4 What test instrument would you use to do an insulation resistance test? (1)

1.5 You have to install THREE lights in parallel in a client's kitchen. The lights are controlled from a single-pole switch.

Draw a neat, labelled circuit diagram of the light circuit. The diagram should include a distribution board showing all the components. (All components in the circuit must be earthed.) (10)

[25]

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QUESTION 2

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

2.1.1 A cable gland is used to attach an armoured cable mechanically to an enclosure.

2.1.2 Line taps can be used when joining overhead lines.

2.1.3 When doing a crimped ferrule joint, side cutters can be used to crimp the joint.

2.1.4 A flexible metal conduit may be used as an earth continuity conductor.

2.1.5 A coupling is used to extend conduits.

(5 × 1) (5)

2.2 You have joined a damaged conductor and must to do a continuity test on it.

2.2.1 What test instrument will you use? (1)

2.2.2 On what scale would you set the test instrument? (1)

2.2.3 Briefly explain how you would conduct the test. (2)

2.2.4 On completion of the test an infinity reading is recorded.

Is this reading acceptable? Motivate your answer. (2)

2.3 Trunking is one of the wireways used in the electrical industry.

List FOUR advantages of trunking over other types of wireways. (4)

2.4 Explain how you would join TWO PVC insulated steel-wire armoured cables using a joining kit. (5)

[20]

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QUESTION 3

3.1 FIGURE 1 below shows the construction of a DC machine.

Identify the parts labelled A–H by writing only the name of the parts next to the letter (A–H) in the ANSWER BOOK.

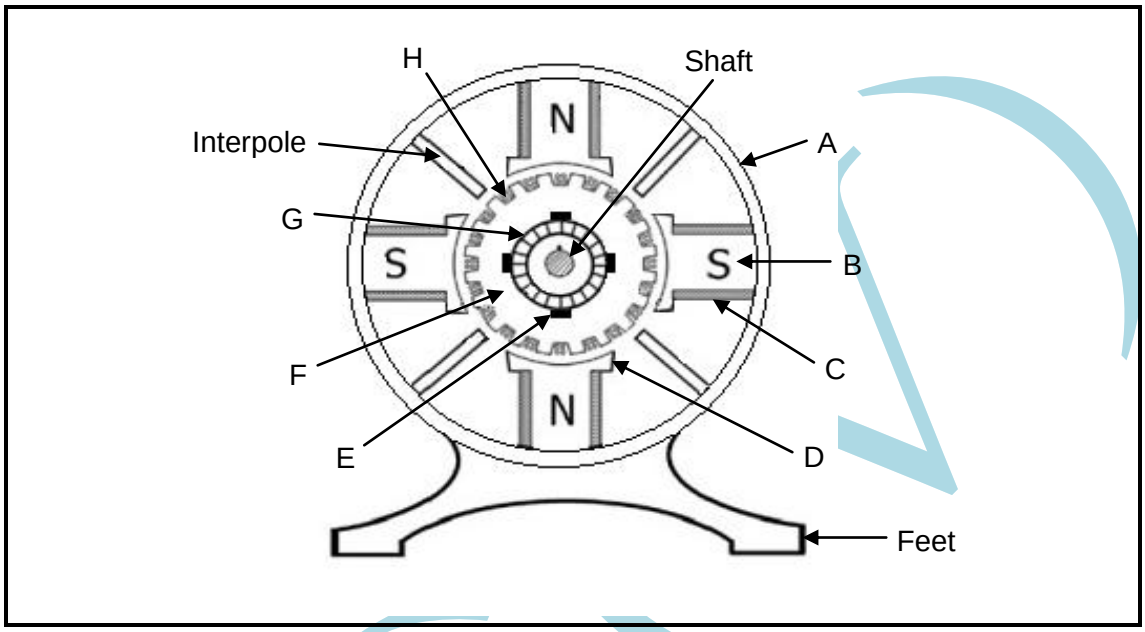


FIGURE 1 (8)

3.2 Interpoles, as shown in FIGURE 1, are small poles mounted inside a DC machine.

What is the function of the interpoles? (2)

3.3 Name THREE mechanical checks that must be done on the shaft of an electrical machine. (3)

3.4 The feet of an electrical machine is used for mounting the machine.

Name TWO types of bases used for mounting these machines. (2)

[15]

QUESTION 4

4.1 An earth leakage relay in a domestic installation needs to be tested regularly to ensure correct operation.

4.1.1 Name TWO tests that must be performed on an earth leakage unit. (2)

4.1.2 What test instrument can be used to perform one of the tests on an earth leakage relay? (1)

4.1.3 While performing this test the earth leakage relay trips at 45 mA.

Is this value acceptable? Motivate your answer. (2)

4.2 Below is a table of tests that can be performed on a circuit breaker. Write only the test instrument you would use to perform the relevant test next to the question number (4.2.1–4.2.5) in the ANSWER BOOK.

TEST PERFORMED	TEST INSTRUMENT
Continuity test	4.2.1
Insulation resistance test	4.2.2
Mechanical operation	4.2.3
Overload tripping test	4.2.4
Contact resistance test	4.2.5

(5)

4.3 A geyser stopped supplying hot water. You are required to test the element to determine whether it is faulty or not.

Explain the procedure you would follow when conducting the test on the element. (Include safety in your explanation.) (5)

[15]

QUESTION 5

5.1 A client wants a geyser to switch off automatically at 07:00 and switch it back on at 16:00.

What energy control device would you recommend him to install? (1)

5.2 The two main types of measuring instruments are recording instruments and testing instruments.

Name TWO types of displays used in these instruments. (2)

5.3 Show, with the aid of a neatly labelled circuit diagram, how an ammeter and a voltmeter are connected in a DC circuit. (7)

5.4 A faulty voltmeter mounted on a panel needs to be replaced.

Make a list of tools you would need to complete this task. (5)

[15]

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QUESTION 6

6.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–G) next to the question number (6.1.1–6.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
6.1.1	Power inverter	A	stores electrical energy
6.1.2	Diode	B	regulates the charging voltage
6.1.3	PV module	C	converts low-voltage DC to AC to run most appliances
6.1.4	Array	D	interconnected photovoltaic cells designed to deliver voltage and current at its output terminals
6.1.5	Battery bank	E	allows current to pass in one direction only
		F	converts electrical energy to light
		G	modules connected together

(5 × 1) (5)

6.2 Study FIGURE 2 below and answer the questions.

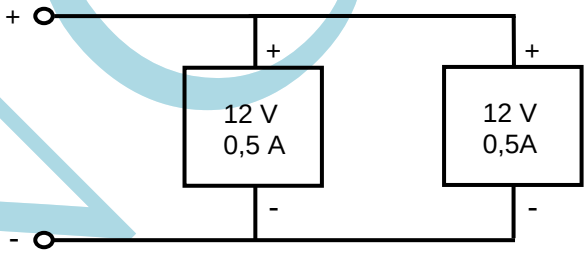


FIGURE 2

6.2.1 How are the TWO photovoltaic (PV) modules connected in the above circuit? (1)

6.2.2 What is the total current for the configuration? (2)

6.2.3 What is the total voltage for the configuration? (2)

[10]

TOTAL: 100

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